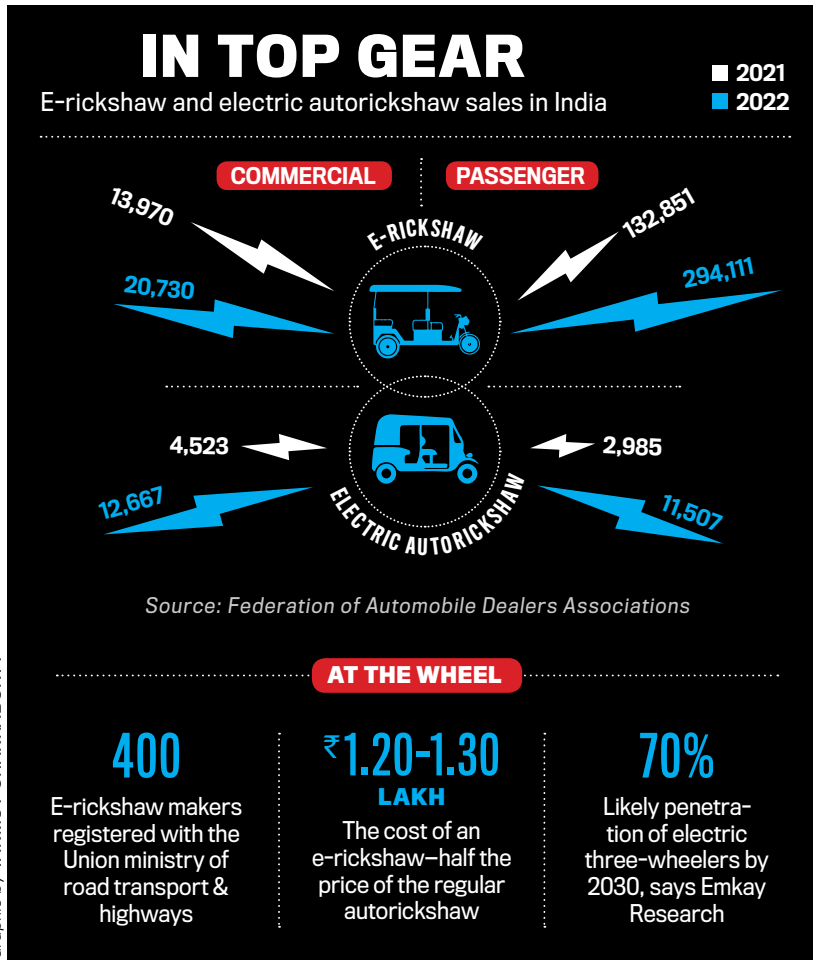


Graphic by TANMOY CHAKRABORTY

Banerjee of Saera Electric, which markets both Li-ion and lead-acid e-rickshaws under the Mayuri brand. From a single plant in Bhiwadi in Rajasthan, Saera acquired a second facility in 2021 at Bawal, Haryana—where Harley-Davidson previously manufactured motorcycles. Now, it is setting up a third plant near Mathura in Uttar Pradesh. All put together, Saera expects to reach a production capacity of 74,000 e-rickshaws this year. It is now gearing up to move into the high-speed electric autorickshaw segment.

Compared to three years ago, there are now more models available in the electric auto space—besides big players such as Mahindra and Piaggio, this segment of green mobility is also attracting new players. "We believe three-wheelers are going to be the game-changer," says Amitabh Saran, founder and CEO of Altigreen Propulsion Labs, a Bengaluru-based electric autorickshaw manufacturer in which Reliance Industries has invested. This January, Altigreen unveiled a cargo variant which claims to recharge its batteries in 15 minutes as against the 3.5 hours that electric autorickshaws usually take for a full recharge. The rapid charging takes place via a battery management system, which—using software and electronics designed by another Bengaluru start-up, Exponent Energy—can funnel power safely into the vehicle's Lithium Iron Phosphate (LFP) battery. Saran plans to set up 100 fast-charging outlets across 12 cities starting with Bengaluru, this year. "Remember the PCO (public call office) booths? Every shop had a pay phone back then. We are saying that charging is going to become a PCO model."

> **"WE BELIEVE THREE-WHEELERS ARE GOING TO BE THE GAME-CHANGER. REMEMBER THE PCO (PUBLIC CALL OFFICE) BOOTHS? CHARGING IS GOING TO BECOME A PCO MODEL"**
>
> **-AMITABH SARAN**
> *Founder and CEO, Altigreen Propulsion Labs, Bengaluru*

In India, the transition to electric vehicles will largely ride on two-wheelers because of the sheer size of that segment—nearly 20 million motorcycles and scooters were sold in India at the industry's peak in FY2018. But the three-wheeler market has a special attraction because of its versatility. It straddles the passenger and the cargo roles—for both purposes, the sweet spot in terms of range is around 100 km per day. Hence, start-ups are eyeing either battery swapping or rapid charging networks.

The penetration of electric three-wheelers is likely to be faster than either two-wheelers or buses, reaching 70 per cent by 2030 in the best-case scenario, according to Emkay Research. Agrees Arun Sreyas, co-founder and CEO of Hyderabad-based start-up RACEnergy, which has passenger autorickshaws firmly in its sights. "Passenger autos are a massive market," he says, pointing to the sheer number of these vehicles on Indian roads, currently running on fuels like LPG or CNG. Over the past year, RACEnergy has been running pilots in Hyderabad and Tiupati to convert these internal combustion-engined vehicles into electric autorickshaws by retrofitting Li-ion battery packs into them. "It is a market that needs to be electrified," he says. "For half a century, autorickshaws haven't changed much." That, he's certain, is about to change. ■